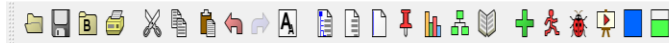


All Files Sort By Name

← → ↺ ↻

C:\Users\Kwadwo Larbi\Desktop\TTTLarbi

```
1  /* Assignment Tic-Tac-Toe
2  Name: Kwadwo Larbi
3  File Created on July 2, 2026
4
5
6  */
7
8  /*
9  BEGIN PROGRAM
10
11  DECLARE board as 2D array of char size 3x3
12  DECLARE currentPlayer as char ('X' for human, 'O' for computer)
13  DECLARE winnerStatus as integer
14
15  CALL initBoard()
16
17  SET currentPlayer = 'X' // human always starts
18
19  DO
20      IF currentPlayer == 'X' THEN
21          CALL yourTurn()
22      ELSE
23          CALL machineTurn()
24      ENDIF
25
26      CALL gameBoard()
27
28      winnerStatus = checkWinner()
29
30      IF winnerStatus == 1 THEN
31          DISPLAY "You win!"
32      ELSE IF winnerStatus == -1 THEN
33          DISPLAY "Computer wins!"
34      ELSE IF winnerStatus == 0 THEN
35          DISPLAY "It's a tie!"
36      ENDIF
37
38      SWITCH currentPlayer:
39          CASE 'X': currentPlayer = 'O'
40          CASE 'O': currentPlayer = 'X'
41      END SWITCH
42
```



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TTTLarbi.class
TTTLarbi.java

```
172 // Main game loop
173 public static void main(String[] args) {
174
175     initBoard();
176     System.out.println("Let's begin!");
177     gameBoard();
178
179     int status = 2;
180     char currentPlayer = 'X';
181
182     while (status == 2) {
183
184         if (currentPlayer == 'X') {
185             yourTurn();
186         } else {
187             machineTurn();
188         }
189
190         gameBoard();
191         status = checkWinner();
192     }
```

TTTLarbi.java

Compile Messages jGRASP Messages Run I/O Interactions

Stop

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```
TTTLarbi.java:175: error: cannot find symbol
    initBoard();
    ^
symbol:   method initBoard()
location: class TTTLarbi
TTTLarbi.java:177: error: cannot find symbol
    gameBoard();
    ^
symbol:   method gameBoard()
location: class TTTLarbi
TTTLarbi.java:185: error: cannot find symbol
        yourTurn();
        ^
symbol:   method yourTurn()
location: class TTTLarbi
TTTLarbi.java:187: error: cannot find symbol
        machineTurn();
        ^
```



All Files Sort By Name



C:\Users\Kwadwo Larbi\Desktop\TTTLarbi

TTTLarbi.class

TTTLarbi.java

```
51 static char[][] board = new char[3][3];
52 static Scanner input = new Scanner(System.in);
53
54
55 // Initialize board
56 public static void initBoard() {
57     for (int r = 0; r < 3; r++) {
58         for (int c = 0; c < 3; c++) {
59             board[r][c] = ' ';
60         }
61     }
62 }
63
64
65 // Display board
66 public static void gameBoard() {
67     for (int r = 0; r < 3; r++) {
68         for (int c = 0; c < 3; c++) {
69             System.out.print("|" + board[r][c]);
70         }
71         System.out.println("|");
72     }
73     System.out.println();
74 }
75
76
77 // Check winner
78 // Return: 1 = human wins, -1 = computer wins, 0 = tie, 2 = no winner yet
79
80 public static int checkWinner() {
81
82     // Check rows
83     for (int r = 0; r < 3; r++) {
84         if (board[r][0] == board[r][1] && board[r][1] == board[r][2] && board[r][0] != ' ') {
85             return (board[r][0] == 'X') ? 1 : -1;
```

TTTLarbi.java *

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```
TTTLarbi.java:175: error: cannot find symbol
    initBoard();
    ^
    symbol:   method initBoard()
    location: class TTTLarbi
TTTLarbi.java:177: error: cannot find symbol
```



All Files Sort By Name

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- TTTLarbi.class
- TTTLarbi.java

```
113     if (full) return 0;
114
115     return 2; // no winner yet
116 }
117
118 // Human turn
119 public static void yourTurn() {
120     int row = -1, col = -1;
121     boolean valid = false;
122
123     while (!valid) {
124         System.out.print("Please enter coordinate of your location (x): ");
125         String move = input.nextLine().trim();
126
127         // Expect format like "1,2"
128         if (move.length() == 3 &&
129             Character.isDigit(move.charAt(0)) &&
130             move.charAt(1) == ',' &&
131             Character.isDigit(move.charAt(2))) {
132
133             row = move.charAt(0) - '0';
134             col = move.charAt(2) - '0';
135
136             if (row >= 0 && row <= 2 && col >= 0 && col <= 2) {
137                 if (board[row][col] == ' ') {
138
```

TTTLarbi.java

Compile Messages jGRASP Messages Run I/O Interactions

End

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Help



Let's begin!

!!!!
!!!!
!!!!

Please enter coordinate of your location (x):

----jGRASP: Process ended by user.

----jGRASP exec: java TTTLarbi

Let's begin!

!!!!
!!!!
!!!!

Please enter coordinate of your location (x):

```

149         System.out.println("Invalid format. Use row,col (e.g., 1,2)");
150     }
151 }
152 }
153
154 // Computer turn
155 public static void machineTurn() {
156     int row, col;
157
158     while (true) {
159         row = (int)(Math.random() * 3);
160         col = (int)(Math.random() * 3);
161
162         if (board[row][col] == ' ') {
163             board[row][col] = 'O';
164             // computer place test
165             // System.out.println("Computer placed O at (" + row + ", " + col + ")");
166             return;
167         }
168     }
169 }
170 }
171
172 // Main game loop
173 public static void main(String[] args) {
174     initBoard();
175     System.out.println("Let's begin!");
176     gameBoard();
177
178     int status = 2;
179     char currentPlayer = 'X';
180
181

```

TTTLarbi.java *

Compile Messages jGRASP Messages Run I/O Interactions

End

Clear

Help

A

<

>

```

! ! ! !
! ! ! !
! ! ! !

```

▶ Please enter coordinate of your location (x): 1,0

```

! ! ! !
!X! ! !
! ! ! !

```

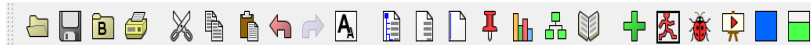
Computer placed O at (0, 1)

```

! !O! !
!X! ! !
! ! ! !

```

▶ Please enter coordinate of your location (x):



All Files Sort By Name



C:\Users\Kwadwo Larbi\Desktop\TTTLarbi

TTTLarbi.class

TTTLarbi.java

```
169     }
170 }
171
172
173 // Main game loop
174 public static void main(String[] args) {
175
176     initBoard();
177     System.out.println("Let's begin!");
178     gameBoard();
179
180     int status = 2;
181     char currentPlayer = 'X';
182
183     while (status == 2) {
184
185         if (currentPlayer == 'X') {
186             yourTurn();
187         } else {
188             machineTurn();
189         }
190
191         gameBoard();
192         status = checkWinner();
193
194         if (status == 1) {
195             System.out.println("You won!");
196         } else if (status == -1) {
197             System.out.println("You lost!");
198         } else if (status == 0) {
199             System.out.println("It's a tie!");
200         }
201     }
```

TTTLarbi.java

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End

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Help



----jGRASP exec: java TTTLarbi

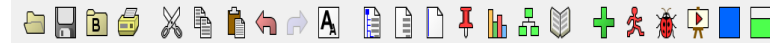
Let's begin!

! ! ! !

! ! ! !

! ! ! !

- ▶ Please enter coordinate of your location (x): 00
- Invalid format. Use row,col (e.g., 1,2)
- ▶ Please enter coordinate of your location (x):



All Files Sort By Name



C:\Users\Kwadwo Larbi\Desktop\TTTLarbi

TTTLarbi.class

TTTLarbi.java

```
169     }
170 }
171
172
173 // Main game loop
174 public static void main(String[] args) {
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176     initBoard();
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185         if (currentPlayer == 'X') {
186             yourTurn();
187         } else {
188             machineTurn();
189         }
190
191         gameBoard();
192         status = checkWinner();
193
194         if (status == 1) {
195             System.out.println("You won!");
196         } else if (status == -1) {
197             System.out.println("You lost!");
198         } else if (status == 0) {
199             System.out.println("It's a tie!");
200         }
201     }
```

TTTLarbi.java

Compile Messages jGRASP Messages Run I/O Interactions

End

Clear

Help



```
!X! !X!
! ! !O!
! ! !O!

>> Please enter coordinate of your location (x): 0,1
!X!X!X!
! ! !O!
! ! !O!

You won!
Press any key to continue...

----jGRASP: Operation complete.
>> [
```